9 Only one isotope of gold occurs naturally on Earth.

(a) State what this indicates about the nuclear structure of all the naturally occurring atoms of gold on Earth.

.....

..... [1]

(b) There are several artificially produced isotopes of gold.

Gold-198 ($^{198}_{79}Au$) is an artificial isotope which is used in medicine and in scientific research.

Gold-198 decays by β (beta)-emission to a stable isotope of mercury.

(i) Determine the number of protons and the number of neutrons in a nucleus of this isotope of mercury.

number of protons = ...

number of neutrons = ...
[2]

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(ii) A sample of gold-198 is placed near to a radiation detector in a research laboratory. The count rate is recorded at the same time every day for 32 days.

The results are used to plot the graph shown in Fig. 9.1.

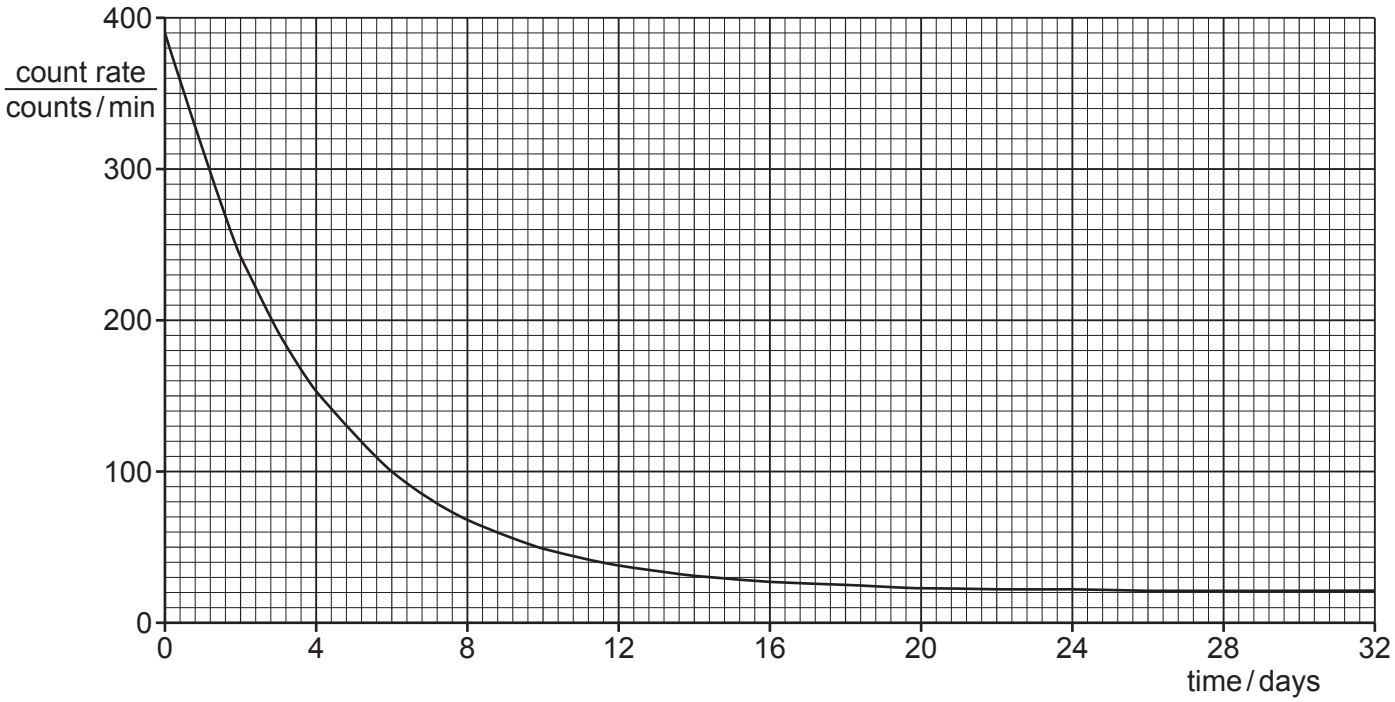


Fig. 9.1

Using Fig. 9.1, determine the background count rate in the research laboratory.

count rate = ... [1]

(iii) Using Fig. 9.1, determine the half-life of gold-198.

half-life = ... [4]

[Total: 8]

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